

Riphah International University

Introduction to Crypto & Blockchain

Project Proposal

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Blockchain-Based Voting System

1. Introduction

The Blockchain-Based Voting System is a web-based project that will allow registered users to vote in an online election. The votes will be recorded using blockchain technology through a smart contract. The main purpose of the project is to create a simple voting system where votes can be recorded securely and the results can be checked easily.

2. Problem Statement

Traditional voting systems can require a lot of paperwork and manual work. In an online voting system, there can also be concerns about changing or deleting voting records. This project will use blockchain to keep voting records in a way that is difficult to change after they are recorded.

3. Proposed Solution

I propose a simple blockchain-based voting application for a university or class election. An administrator will create an election, add candidates and register voters. A registered voter will connect a digital wallet and cast one vote. The smart contract will record the vote on the blockchain and the application will display the final results.

4. Objectives

- To develop a simple online voting system using blockchain technology.
- To use a smart contract for recording votes.
- To make sure a registered voter can vote only once.
- To allow an administrator to manage the election and candidates.
- To display voting results clearly.
- To understand the practical use of blockchain and smart contracts.

5. Main Features

- Admin can create and manage an election.
- Admin can add candidates.
- Admin can register eligible voters.

- Voters can connect their MetaMask wallet.
- Voters can cast one vote.
- The smart contract records votes on the blockchain.
- The system prevents a voter from voting more than once.
- Users can view the election results after voting ends.

6. Technology to be used

- Solidity - for developing the smart contract.
- Ethereum - for testing the project.
- MetaMask - for connecting the user's wallet.
- Ethers.js - for connecting the website with the smart contract.
- HTML, CSS and JavaScript - for the website interface.

7. How the System Will Work

First, the administrator will create an election and add the candidates. Eligible voters will then be registered. A voter will connect MetaMask to the website and select a candidate. The smart contract will check whether the voter is registered and has already voted. If everything is valid, the vote will be recorded on the blockchain. After the election ends, the system will show the number of votes received by each candidate.

8. Expected Outcome

At the end of the project, I expect to have a working web application that can conduct a small university or class-level election. The project will demonstrate how blockchain and smart contracts can be used to record and verify voting transactions.

9. Project Scope

The project will be developed as an educational prototype for a university or classroom election. It will not be designed for national or government elections. The main focus will be on learning and demonstrating blockchain concepts.

10. Conclusion

The Blockchain-Based Voting System will provide a simple practical example of how blockchain can be used in voting. The project will help students understand smart contracts, blockchain transactions, digital wallets and decentralized applications.